AERONEX PRECISION COMPONENTS PVT. LTD. | Manufacturing Operations Analytics

Business Background

Aeronex Precision Components Pvt. Ltd. is a leading manufacturer of high-precision industrial components supplying automotive, aerospace, and heavy engineering companies across India.

The company operates multiple production lines with automated machinery, skilled operators, and strict quality control processes. Despite significant investments in manufacturing infrastructure, management has observed variations in production efficiency, recurring machine downtime, inconsistent operator performance, and increasing quality defects.

Operational data currently exists across multiple systems, making it difficult for plant managers and leadership teams to monitor performance and identify improvement opportunities. The Operations Excellence team requires a centralized Power BI reporting solution to measure production performance, machine utilization, downtime trends, and product quality metrics.

Problem Statement

You are a Junior Data Analyst assigned to Aeronex Precision Components Pvt. Ltd.

Starting from the raw multi-table dataset, your task is to clean and transform the data, create an optimized data model, develop analytical KPIs using DAX, and build an interactive Power BI dashboard that enables management to monitor operational performance and identify areas for process improvement.

Dataset Tables & Fields Used

1. Production Log
2. Machine Master
3. Operator Master
4. Product Master
5. Downtime Log
6. Quality Inspection Log
7. Maintenance Records
8. Shift Calendar
9. Data Dictionary

Key Tasks to Perform

- Analyse all eight datasets for missing values, duplicate records, incorrect data types, and inconsistent naming conventions; perform necessary cleaning and transformations using Power Query.

- Create an optimised data model by establishing appropriate relationships between all datasets and implementing a proper star schema design.

- Develop DAX measures to calculate Production Achievement %, Machine Utilisation %, Downtime %, Defect Rate %, and Overall Equipment Effectiveness (OEE).

- Build an executive KPI dashboard displaying Total Production, Total Downtime, Machine Utilisation %, Defect Rate %, and OEE.

- Create interactive visualisations to analyze production performance across plants, machines, operators, shifts, and product categories.

- Analyze downtime trends and identify the machines, shifts, and downtime reasons contributing most to production losses.

- Evaluate quality performance by identifying defect patterns, high-risk products, and operational areas requiring corrective action.

- Prepare an operations performance report highlighting the best-performing machine, most efficient operator, highest downtime contributor, highest defect-producing product category, and key recommendations for improving operational efficiency.

Final Deliverables

- Data Audit Report
- Cleaned & Transformed Datasets
- Power BI Data Model
- DAX Measures & KPIs
- Interactive Operations Dashboard
- Production & Downtime Analysis
- Quality Performance Analysis

Evaluation Criteria

Criteria	Weightage
Data Cleaning & Transformation	20%
Data Modeling & Relationships	20%
DAX Measures & KPIs	25%
Dashboard Development	20%